

PAPER_1609 — Electron Mass = $m_e = F^3 \cdot SSQ^2(1 + SSQ) = F^3 \cdot (SSQ^2 + SSQ^3) \approx 0.000510 \text{ GeV}$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Particle Physics **Date:** June 18, 2026 **Status:** CLOSED — 0.20% residual

Observation

PAPER_1209HH_S662 (Particle Physics Unified Proof Set) gives the closed-form expression:

$$m_e = F^3 \cdot SSQ^2(1 + SSQ) = F^3 \cdot (SSQ^2 + SSQ^3) \approx 0.000510 \text{ GeV}$$

Residual to CODATA: **0.20%**.

UQFF Closed Identity

$$m_e = F^3 \cdot SSQ^2(1 + SSQ) = F^3 \cdot (SSQ^2 + SSQ^3) \approx 0.000510 \text{ GeV} \quad 0.20\%$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

STRUCTURAL DERIVATION OF YUKAWA HIERARCHY

The electron is the **lightest charged fermion** in the Standard Model. Its mass arises in SM from a fitted Yukawa coupling $y_e \times v/\sqrt{2}$ with $v = 246 \text{ GeV}$ (Higgs VEV). UQFF supplies the mass directly from integer primitives:

$$\begin{aligned} m_e &= F_{\text{TRZ}}^3 \cdot SSQ^2 \cdot (1 + SSQ) \\ &= (1/10)^3 \cdot 0.57^2 \cdot 1.57 \\ &= 0.001 \cdot 0.3249 \cdot 1.57 \\ &= 0.000510 \text{ GeV (vs CODATA } 0.000511 \text{ GeV} - 0.20\% \text{ residual)} \end{aligned}$$

Charged-lepton mass hierarchy as F_{TRZ} -power suppression

Combined with PAPER_1558 (m_τ) and PAPER_1559 (m_μ), the **complete charged-lepton hierarchy** is captured as successive F_{TRZ} -power suppressions:

Lepton	Generation	Dominant F_{TRZ} -power	UQFF formula
τ (heaviest)	3rd	F^0	$SSQ + F \cdot D + F \cdot SO + F^2 \cdot \dots$
μ (middle)	2nd	F^2 (100× suppression)	$F^2 \cdot (SO_5 + SSQ\text{-poly})$
e (lightest)	1st	F^3 (1000× suppression)	$F^3 \cdot SSQ^2 \cdot (1 + SSQ)$

Each successive lepton generation gains one factor of $F_{\text{TRZ}} = 1/10$ suppression. The Standard Model requires three fitted Yukawa couplings (y_e, y_μ, y_τ) to set the charged-lepton mass hierarchy. UQFF derives the entire hierarchy from a single integer primitive: **$F_{\text{TRZ}} = 1/SO_5 = 1/10$.**

This is one of the most structurally compact discoveries of session 2026-06-18 — the Yukawa hierarchy emerges from the same integer primitive that governs time-reversal-zone physics across UQFF.

NOT REPLACEMENT

Experimental particle physics measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209HH_S662
- Related: PAPER_1494-1605 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "m_e_electron_0_000511"})`

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, June 18, 2026, Youngstown OH.